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Homoenergetic solutions to the Landau equation: well-posedness vs hydrodynamic blow-up

joint with M.-P. Gualdani and A. Nota

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The Landau equation

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The **Landau equation** reads:

$$(\partial_t + v\cdot\nabla_x) f = \nabla_v \cdot \int_{\mathbb{R}^3} |v-v_*|^\gamma \Phi(v-v_*)\nabla_{v-v_*} f(x,v) f(x,v_*)\, dv_*,$$

where $\Phi(z) = |z|^2 I - z\otimes z$, and $\gamma \geq -3$. State of the art:

Value of γ	$[-3,-2]$	$(-2,1)$	$(\sqrt{3},2]$
Global existence space homogeneous	✓ [GS,'25]	✓ [GS,'25]	✓ [GS,'25]
Global existence near equilibrium	✓ [G,'02]	✓ [G,'02]	✓ [G,'02]
Global existence near vacuum	?	✓ [L,'19], [C,'23]	?
Finite-time singularities	?	?	✓ [BCGJVY,'26]

Homoenergetic solutions

Recall the macroscopic quantities $\rho(t,x)$, $u(t,x)$, $e(t,x)$

$$\rho = \int f\,dv, \qquad \rho u = \int vf\,dv, \qquad \rho e = \int |v-u|^2 f\,dv.$$

Main feature: $\rho(t,x) = \rho(t)$, $e(t,x) = e(t)$, $u(t,x) = L(t)x$.

Derived from the ansatz

$$f(t,x,v) = g(t,w), \qquad w = v - \xi(t,x),$$

which removes the variable x when $\xi(t,x) = L(t)x$, with

$$L'(t) = -L(t)^2 \implies L(t) = A(I+tA)^{-1}.$$

Properties of initial data: $f_{\text{in}}(x,v) = g_{\text{in}}(v - Ax)$.

- We have $f_{\text{in}} \in L^\infty_x L^1_v \setminus L^1_{x,v}$.
- There is **no** $C, m > 0$ such that $f_{\text{in}}(x,v) \le C|v|^{-m}$.

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Global well-posedness

Theorem 1: Existence of smooth solutions

Let $\gamma \geq -3$ and $A \in \mathbb{R}^{3\times 3}$ with $\det(I + tA) > 0$ for $0 \leq t \leq T$. Let

$$f_{\text{in}}(x, v) = g_{\text{in}}(v - Ax), \qquad g_{\text{in}} \in \mathcal{S}(\mathbb{R}^3), \quad g_{\text{in}} \geq 0.$$

Then, there exists a classical solution $f \in \mathcal{C}^\infty\left([0, T] \times \mathbb{R}^6\right)$ with

$$f(t, x, v) = g(t, v - L(t)x), \qquad L(t) = A(I + tA)^{-1},$$

where $g(t, w)$ satisfies $\partial_t g = \langle L(t)w, \nabla_w g\rangle + Q_w(g)$.

Main ingredients:

- Boundedness of Fisher information $\displaystyle\int \frac{|\nabla g|^2}{g}\, dw.$
- Moment estimates of g and coercivity of $|w|^\gamma \Phi(w) * g$.

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Theorem 2: Hydrodynamic blow-up for $A = -I$

Let $\gamma \geq -3$, and initial data

$$f_{\text{in}}(x, v) = h_{\text{in}}(x + v), \qquad h_{\text{in}} \in \mathcal{S}(\mathbb{R}^3), \quad h_{\text{in}} \geq 0.$$

Then there is $f \in \mathcal{C}^\infty\left([0,1] \times \mathbb{R}^6\right) \cap L^\infty([0,1] \times \mathbb{R}^6)$ with

$$\rho(t) = (1-t)^{-3} \int_{\mathbb{R}^3} g_{\text{in}}\, dw, \qquad e(t) = (1-t)^{-2} \frac{\int_{\mathbb{R}^3} |w|^2 g_{\text{in}}\, dw}{\int_{\mathbb{R}^3} g_{\text{in}}\, dw}.$$

Moreover, f is written as

$$f(t, x, v) = h\left(s(t), (1-t)v + x\right), \quad s(t) := \int_0^t (1-\tau)^{-3-\gamma}\, d\tau,$$

where $h(s, w)$ solves the **homogeneous** Landau equation.

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